

Assignment No.5

Problem Statement : Unnamed PL/SQL code block: Use of Control structure and Exception handling is mandatory. Suggested Problem statement: Consider Tables: 1. Borrower(Roll_no, Name, DateofIssue, NameofBook, Status) 2. Fine(Roll_no,Date,Amt)

Program :

```
CREATE TABLE Borrower ( Roll_no NUMBER(15), Name VARCHAR(255), DateOfIssue DATE, NameOfBook VARCHAR(255), Status VARCHAR(255), PRIMARY KEY(Roll_no) ); Table created.
```

```
CREATE TABLE Fine ( Roll_no NUMBER(15), DateOfReturn DATE, Amount NUMBER(15), FOREIGN KEY(Roll_no) REFERENCES Borrower(Roll_no) ); Table created.
```

```
INSERT INTO Borrower (Roll_no, Name, DateOfIssue, NameOfBook, Status) values (1, 'Ayush Kalaskar', TO_DATE('2024-07-28','YYYY-MM-DD'), 'DBMS', 'I');
```

1 row(s) inserted.

```
INSERT INTO Borrower (Roll_no, Name, DateOfIssue, NameOfBook, Status) values (2, 'Himanshu Patil', TO_DATE('2024-07-01','YYYY-MM-DD'),'IOT', 'I');
```

1 row(s) inserted.

```
INSERT INTO Borrower (Roll_no, Name, DateOfIssue, NameOfBook, Status) values (3, 'Kshitij', TO_DATE('2024-08-05','YYYY-MM-DD'), 'TOC', 'I');
```

1 row(s) inserted.

```
DECLARE
```

```
p_nameofbook VARCHAR(255);
```

```
p_rollno NUMBER(15) := 1;
```

```
p_dateofissue DATE;
```

```
currentdate DATE;
```

```
noofdays NUMBER(15);
```

```
amount NUMBER(15);
```

```
nodata EXCEPTION;
```

```
BEGIN
```

```
p_nameofbook := '&p_nameofbook';
```

```

currentdate := SYSDATE;

IF p_rollno <= 0 THEN RAISE nodata;

END IF;

SELECT DateOfIssue INTO p_dateofissue FROM Borrower WHERE Roll_no = p_rollno
AND NameOfBook = p_nameofbook;

SELECT TRUNC(SYSDATE) - p_dateofissue INTO noofdays FROM dual;
DBMS_OUTPUT.PUT_LINE('No of days: ' || noofdays);

IF (noofdays > 30) THEN amount := noofdays * 50; ELSIF (noofdays >= 15 AND noofdays
<= 30) THEN amount := noofdays * 5; ELSE amount := 0; END IF;

IF amount > 0 THEN INSERT INTO Fine (Roll_no, DateOfReturn, Amount) VALUES
(p_rollno, SYSDATE, amount);

END IF;

IF amount > 0 THEN UPDATE Borrower SET Status = 'R' WHERE Roll_no = p_rollno;
END IF;

EXCEPTION WHEN nodata THEN DBMS_OUTPUT.PUT_LINE('Roll number or book
name not found.');
```

```

WHEN NO_DATA_FOUND THEN DBMS_OUTPUT.PUT_LINE('No entry found for the
provided roll number and book name.');
```

```

WHEN OTHERS THEN DBMS_OUTPUT.PUT_LINE('An error occurred: ' || SQLERRM);

END;/
```

Output :

1. For ROLL_NO = 1 and NAMEOFBOOK = 'DBMS'
Statement processed.

No of days: 63

Borrower:

ROLL_NO	NAME	DATEOFISSUE	NAMEOFBOOK	STATUS
1	Ayush Kalaskar	28-JUL-24	DBMS	R
2	Himanshu Patil	01-JUL-24	IOT	I
3	Kshitij	05-AUG-24	TOC	I

Fine:

ROLL_NO	DATEOFRETURN	AMOUNT
1	29-SEP-24	3150